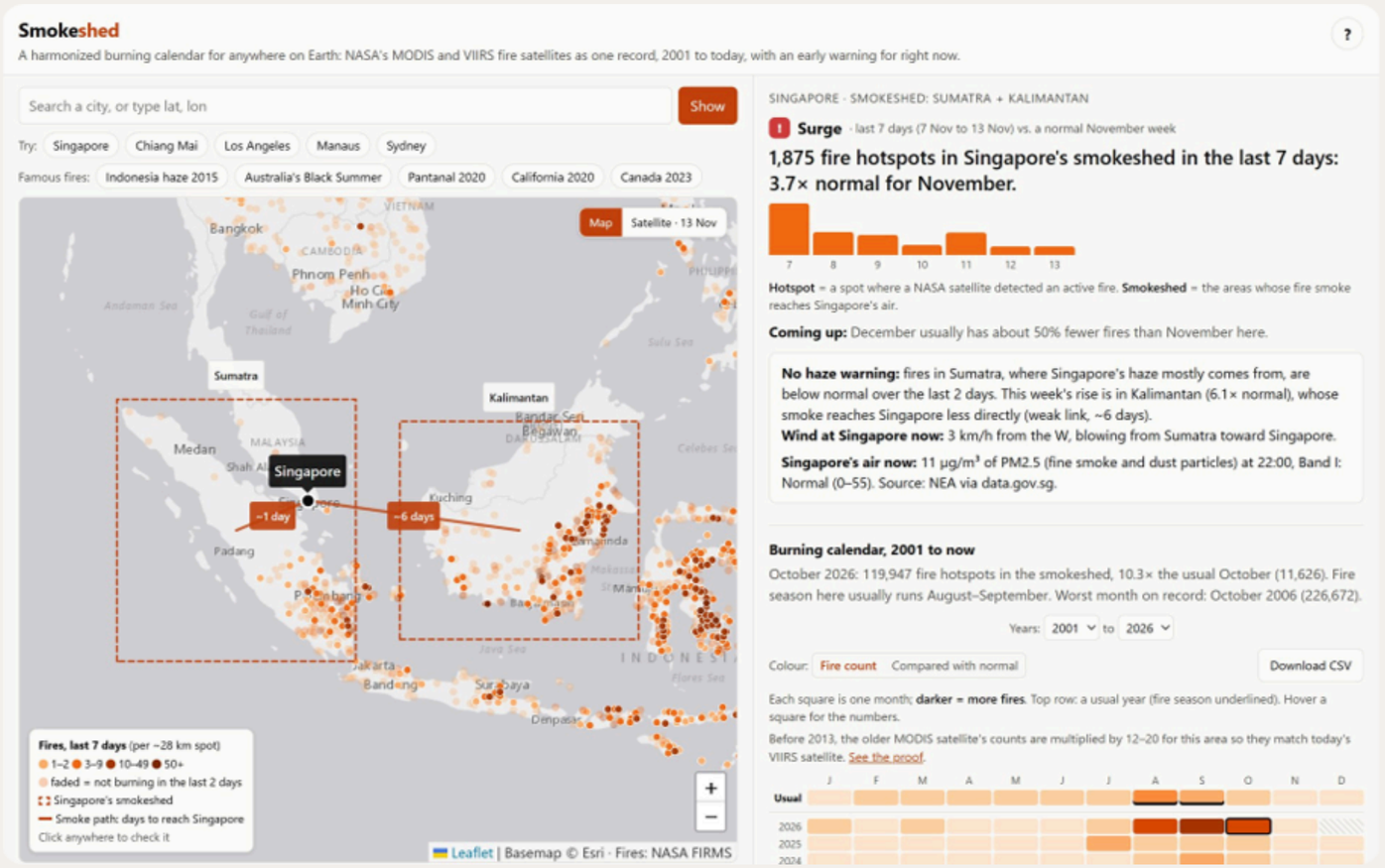
NASA SPACE APPS CHALLENGE 2026

Smokeshed

A harmonized burning calendar for anywhere on Earth, with an early warning for the people who breathe the smoke.

Team members: Tan I-Je

Challenge: Harmonization of MODIS and VIIRS Hot Spots



THE PROBLEM

A Fire Record Split Into Two

1 km

MODIS, since 2000: the long history, but in coarse pixels.

12×

VIIRS, since 2012, sees about 12× as many fires. The two can’t be compared directly.

2026

From late 2026 the MODIS satellites start shutting down. Without harmonizing, 20 years of history lose their meaning.

Fire maps ask “what is burning here?”, never “whose smoke am I breathing?”

WHAT WE BUILT

One Click, One Answer

1 Pick a place

Search, tap a city or a famous fire, or click anywhere on the map.

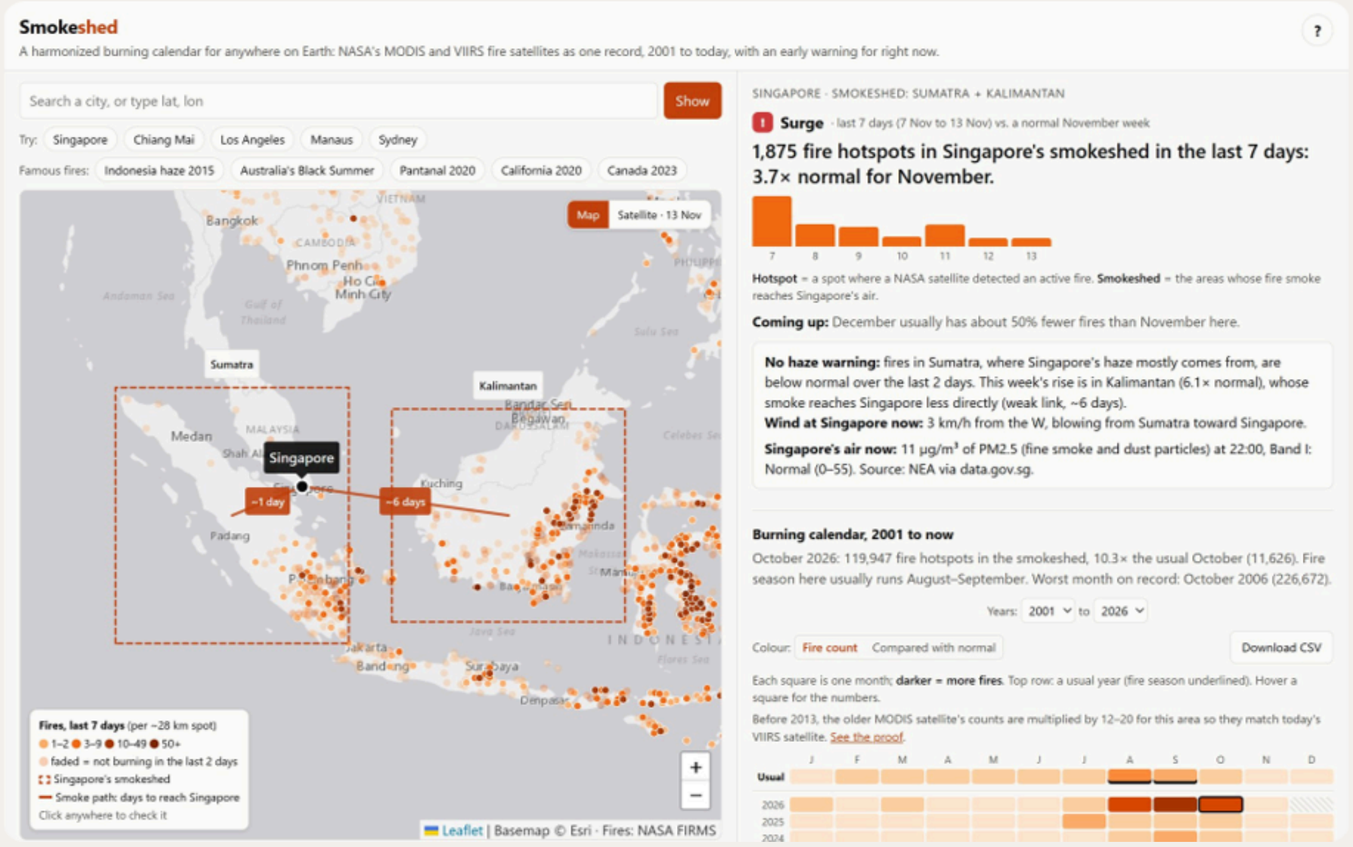
2 See right now

The last 7 days against a normal week: Quiet, Normal, Above normal or Surge.

3 Explore the history

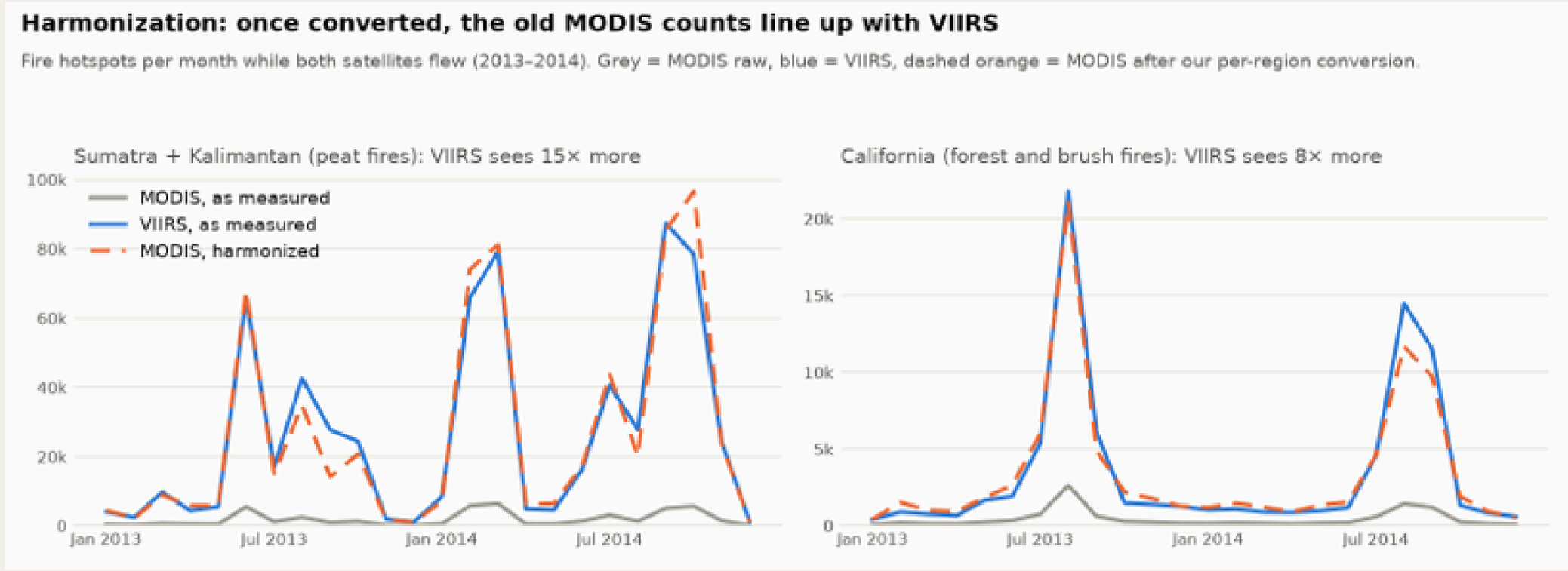
A harmonized calendar since 2001, compared with normal, with a CSV download.

NASA FIRMS, Global Forest Watch and Harvard's SMOKE Policy Tool each hold a piece. Smokeshed is the first to bring them together.



DOES THE HARMONIZATION HOLD UP

Old And New Satellites Hold Up



Indonesia 2015 · Australia 2019–20 · Pantanal 2020 · California 2020 · Canada 2023

11%

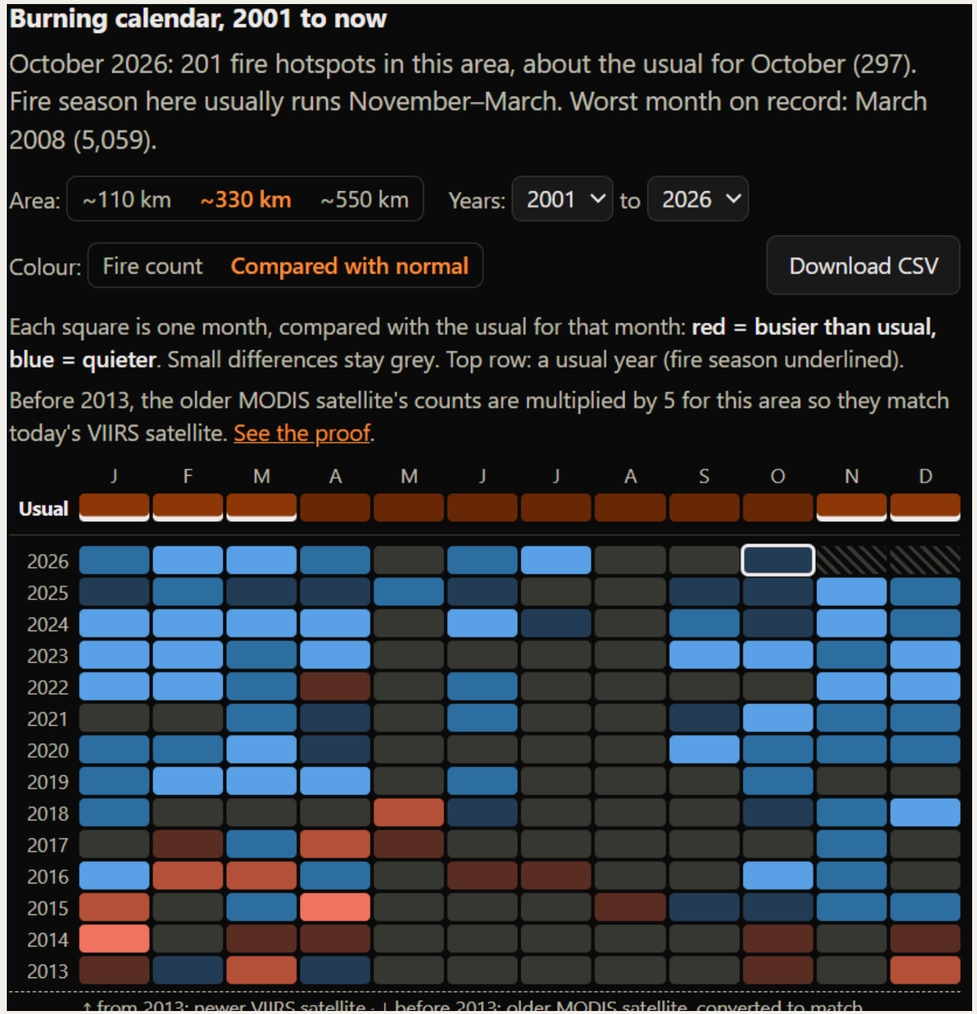
median error when predicting a year the conversion never saw

5 of 5

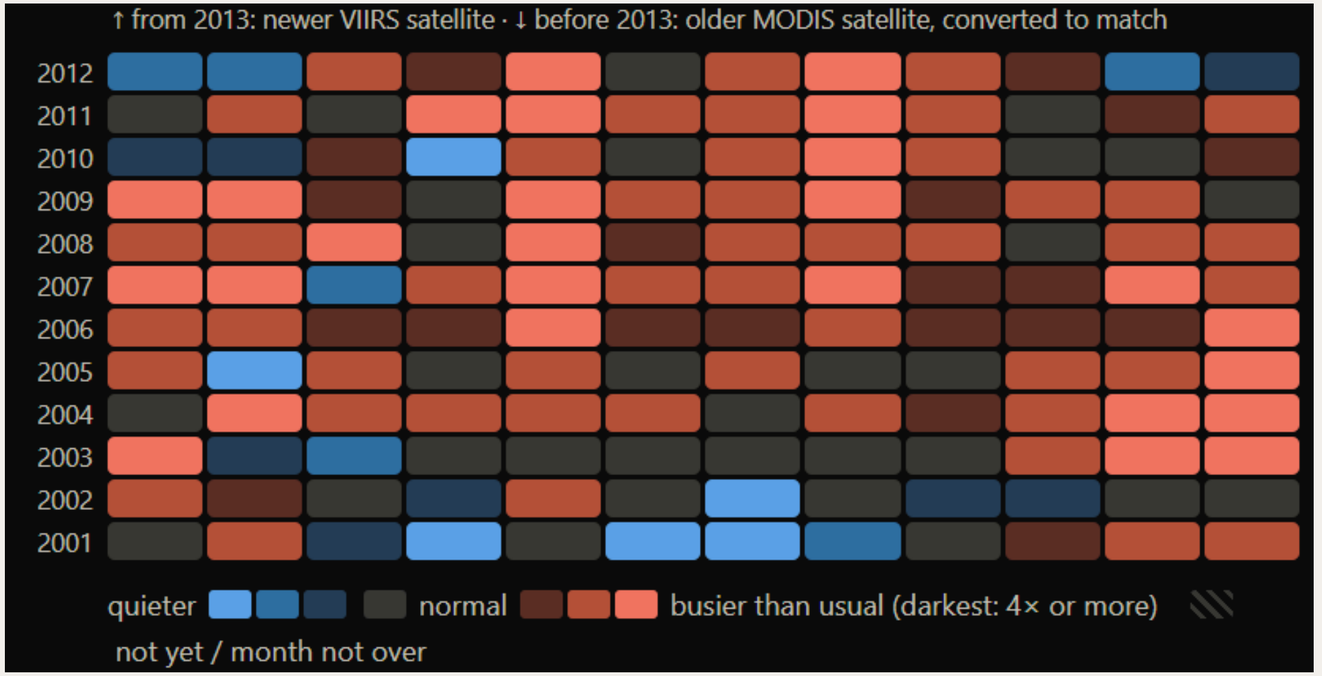
famous fires come out as the #1 month on record where they happened

UNUSUAL CONDITIONS, CRITICAL RECORDS

See Which Years Broke The Pattern



- **Red** = busier than usual for that month; **blue** = quieter.
- South Sumatra: every El Niño drought year turns red (2006, 2015, 2019, 2023), and the wet La Niña years turn blue.
- The top row is a usual year, with the fire season underlined.
- “Coming up” says whether next month is usually busier, so responders can plan ahead.



FOR THE PEOPLE WHO BREATHE THE SMOKE

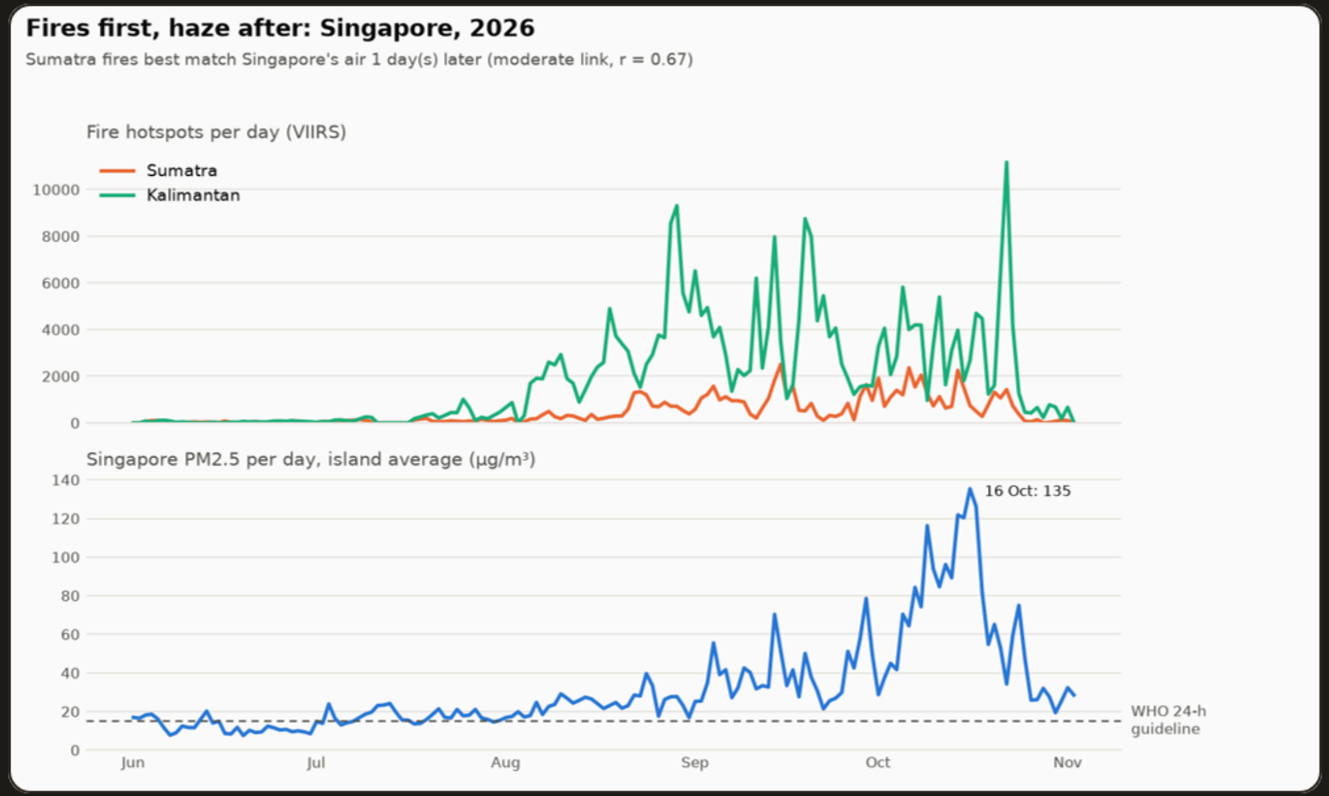
See Which Years Broke The Pattern

~1 day

After Sumatra’s fires surge, Singapore’s air gets worse about a day later (2026 season, r = 0.67).

Singapore has almost no fires of its own. Its **smokeshed** is Sumatra and Kalimantan.

Live wind and Singapore’s air quality turn this into a plain warning, such as “today’s wind is blowing from Kalimantan toward Singapore.”



OPEN NASA DATA, TRY IT OUT!

Built on NASA data

NASA data

- FIRMS MODIS (Terra) fire hotspots, 2001–2014
- FIRMS VIIRS S-NPP 375 m hotspots, 2013–2026
- VIIRS NOAA-20 near real time, refreshed daily
- GIBS true-colour satellite imagery

Also used

- Singapore PM2.5 (NEA, via data.gov.sg)
- Open-Meteo wind
- Esri basemap and OpenStreetMap search
- Python, Leaflet and plain HTML: no server

Next

- Smokesheds for more cities: Kuala Lumpur, Bangkok, Chiang Mai
- Finer grid for land managers
- Where-will-the-smoke-go forecasts
- Alerts for responders

Try it: undonelaundry.github.io/Smokeshed · Code: github.com/UndoneLaundry/Smokeshed